Bayesian Optimization-Based Efficient Impedance Controller Tuning for Robotic Interaction With Force Feedback

Jie Chen, Linsong Deng, Zhengchun Hua, Weiyi Ying, and Jie Zhao, *Senior Member, IEEE*

***Abstract*— Impedance control is widely used in manipulation tasks of robots for the purpose of modulating the contact force between the robot and the interaction environment. Currently, it is achievable to adapt the impedance control parameters to various interactive objects via optimization or learning approaches. However, the adapt efficiency is still one of the most important unsolved challenges. In this regard, this article presents a Bayesian optimization (BO)-based efficient tuning strategy for the impedance controller. The interaction process from impedance control parameters to the produced contact force is modeled as a Gaussian process and the expected improvement plus (EIP) acquisition function is employed as a sampling criterion to tune the impedance control parameters. For interacting with objects, the efficiency and performance of the proposed method are numerically verified and compared with two classical adaptive impedance control schemes. With a real-world robotic manipulator, it is demonstrated that thanks to the inherent data efficiency of the BO, our method is able to find suitable impedance parameters via as few as a dozen of interaction attempts. Finally, the robustness of the proposed method against prior data is discussed.**

***Index Terms*— Controller tuning, efficient tuning strategy, impedance control, real-world robotic interaction.**

I. Introduction

OVER the last few decades, realistic applications (e.g., assembly, medical surgery, disaster response, etc.) are bringing robots into unstructured environments where they are expected to perform complex manipulation tasks [1], [2], [3]. Meanwhile, the interaction between robots and the environment has garnered increasing attention. Under the prerequisite of ensuring safety, robots executing interaction tasks often need to proficiently utilize contact forces to adapt to the environment.

To solve this problem, robots are often built using compliant elements, such as soft materials or springs, which enable safer operation modes [4]. In addition to the robots with built-in compliance, robots can also employ variable stiffness/impedance actuators to adjust their motions or compliance to avoid large impact forces by enforcing desired impedance behaviors by means of control [5].

As a key technique to fulfill the above goal, impedance control was initially proposed by Hogan [6] and later expanded by several authors, e.g., see [7], [8]. This control method offers the advantage of being able to modulate the impedance of the robot, which together with proper tuning can improve efficiency and robustness. Therefore, impedance control has been employed for a wide range in the field of robotics.

Choosing the desired impedance parameters plays an essential role in the actual effect of impedance control (e.g., the control performance of contact force, etc.). The sensitivity of the impedance controller to changes in task settings poses a significant challenge in computing the impedance parameters. As a result, using fixed offline-computed control parameters is rarely a good choice, and thus, the approach of dynamically tuning impedance control parameters becomes popular. For example, in [9], an adaptive variable impedance control method is proposed, which adaptively adjusts the target stiffness for a wheel-legged robot to minimize the force-tracking error of each leg. In [10], a variable impedance controller is developed for an exoskeleton robot for gait rehabilitation. It uses the concept of energy as the basis for tuning gains and integrates a strength metric into its adaptation law. In [11], a fuzzy adaptive impedance control strategy is presented to control the foot force of a hexapod robot. Parameters of the adaptive impedance control are adjusted by the fuzzy controller to adapt to different environments. In [12], an indirect adaptive impedance control strategy is proposed, which calculates the required reference position by estimating environmental parameters online and uses it to control the target contact force.

In recent years, some researchers also attempted learning-based variable impedance control methods. For instance, in [13], a neural network is applied to regulate the end-effector impedance for minimizing position/force control error. However, it is difficult to design the neural network framework and test the number of nodes to guarantee network convergence. Stulp et al. [14] proposed a model-free reinforcement learning method PI2. A seven-DOF simulated robot requires more than 1000 interactions with the environment to obtain the ideal impedance control

Manuscript received 11 July 2023; revised 7 September 2023; accepted 20 September 2023. Date of publication 6 October 2023; date of current version 18 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 52175004 and in part by the Fundamental Research Funds for Central Universities under Grant N2203013. The Associate Editor coordinating the review process was Dr. Shiwu Zhang. *(Corresponding author: Weiyi Ying.)*

Jie Chen, Linsong Deng, Zhengchun Hua, and Weiyi Ying are with the School of Mechanical Engineering and Automation, Northeastern University, Shenyang 110000, China (e-mail: chenjie@me.neu.edu.cn; 2000323@stu.neu.edu.cn; 2100336@stu. neu.edu.cn; 2200450@stu.neu.edu.cn).

Jie Zhao is with the State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150001, China (e-mail: jzhao@hit.edu.cn).

Digital Object Identifier 10.1109/TIM.2023.3322504

1557-9662 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

parameters. Based on PI2, Winter et al. [15] proposed a C-PI2 reinforcement learning algorithm. This algorithm improves the efficiency of PI2 and requires about 120 iterations. In [16], the covariance matrix adaptation evolution strategy (CMA-ES) algorithm is explored to learn gain schedules for variable impedance controller of robot grasping system. After 50 iterations, all the samples of control parameters gradually converge to the desired values. However, in real-world robotic interaction applications, a large number of interaction attempts may cause irreversible physical damage or danger to the robot body. Furthermore, collecting a large amount of data also requires a lot of time and energy. Therefore, the robot needs to adjust the impedance control parameters efficiently to achieve a rapid response to the environment.

From this perspective, this article seeks to investigate an efficient tuning strategy for the impedance controller based on Bayesian optimization (BO), to achieve a force response that meets performance metrics with fewer adjustment times to the impedance controller. Bayesian optimization is a very effective global optimization algorithm for solving complex optimization problems with unknown, on-convex, multimodal, and costly evaluation of the objective function expression [17]. Moreover, Bayesian optimization has been used in the robotic community in the exploration of, for example, gait optimization [18] and hardware design [19]. But few previous works have integrated Bayesian optimization into the impedance control framework for robotic systems. The main contributions of this work are as follows:

1) We propose the introduction of Bayesian optimization into the impedance controller, for automatic and efficient tuning of controller parameters according to the real-time feedback measurement. A complete framework of Bayesian optimization-based impedance control for robotic interaction is developed.
2) The tuning efficiency and performance of the proposed method are numerically evaluated and compared against two benchmark adaptive impedance control schemes over multiple interactive conditions. Our method exhibits an obvious advantage in adaptation efficiency, capable of obtaining appropriate impedance control parameters with fewer interaction attempts.
3) A robotic manipulator consisting of two rotary joints is developed. With the real-world experimental evaluation, it is demonstrated that our method can produce suitable impedance parameters via as few as a dozen of interaction attempts.

II. Method

A. System Description and Control Frame

In the system under study, the end-effector of the robot physically interacts with the environment. The dynamic model of the rigid-joint robotic system with n DOFs in the joint space is given by

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau + J^T(q)F \tag{1}$$

where q, $\dot{q}$, and $\ddot{q}$ denote the position, velocity, and acceleration of the robot joint, respectively, $M(q) \in \mathbb{R}^{n\times n}$ denotes the inertia matrix, $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ is the Coriolis and Centrifugal matrix, $G(q) \in \mathbb{R}^n$ denotes the gravitational term, $\tau \in \mathbb{R}^n$ is the input torque vector, $F(t) \in \mathbb{R}^n$ denotes the interaction force between the robotic system and the environment, and $J^T(q)$ denotes the transpose of the Jacobian matrix.

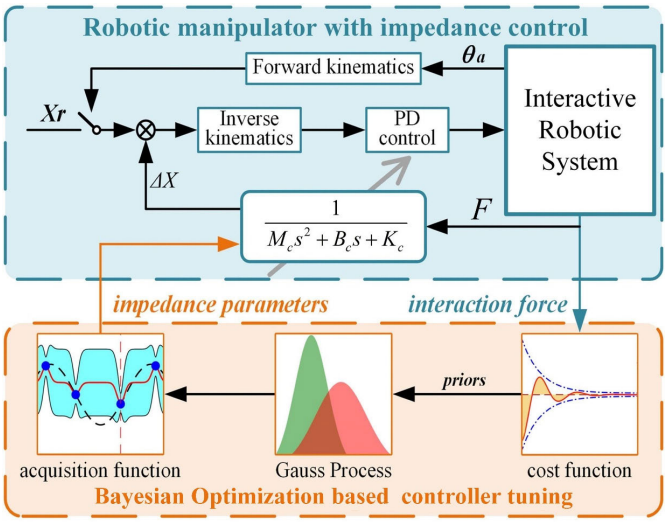


Fig. 1. Block diagram of the proposed Bayesian optimization-based efficient impedance controller tuning for the interactive robotic system.

The other part of the interaction system is the environment. We model the environment as the mass-damping-stiffness system in this article [20]. The environment dynamics are described by the following model:

$$M_e(\ddot{x} - \ddot{x}_e) + C_e(\dot{x} - \dot{x}_e) + G_e(x - x_e) = -F \tag{2}$$

where x and $\dot{x}$ denote the position and velocity of the Cartesian space along the direction of the force, respectively, and x_e is the position of the contact object. In this article, we assume that the position is unknown and varying.

In practical applications, there are inevitably measured errors caused by environmental variations, which make the precise values of the parameter matrices difficult to obtain. Therefore, the parameters M_e, C_e, and G_e in (2) are represented by the known nominal parts M_{e0}, C_{e0}, and G_{e0}, and the bounded uncertain parts ΔM_e, ΔC_e, and ΔG_e, respectively. The environmental dynamics (2) can be rewritten as

$$M_{e0}(\ddot{x} - \ddot{x}_e) + C_{e0}(\dot{x} - \dot{x}_e) + G_{e0}(x - x_e) + f = -F \tag{3}$$

and f is the uncertainty of the dynamic model

$$f(\Delta M_e, \Delta C_e, \Delta G_e) = \Delta M_e(\ddot{x} - \ddot{x}_e) + \Delta C_e(\dot{x} - \dot{x}_e) + \Delta G_e(x - x_e). \tag{4}$$

To adapt to the interactive environment efficiently, a typical position-based impedance control is implemented in the robotic system [21], as shown in Fig. 1. From the perspective of the robotic system

$$M_c(\ddot{x} - \ddot{x}_e) + B_c(\dot{x} - \dot{x}_e) + K_c(x - x_e) = F \tag{5}$$

where M_c, B_c, and K_c denote the desired mass, damping, and stiffness parameters that are chosen to provide the impedance

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

of the end-effector, respectively. Combing (3) and (5) results in the dynamics of the whole system

$$(M_c + M_{e0} + \Delta M_e)\ddot{X} + (B_c + C_{e0} + \Delta C_e)\dot{X} + (K_c + G_{e0} + \Delta G_e)X = 0 \tag{6}$$

where $X = x - x_e$, and the whole system (6) is a typical 2nd-order system, whose damping ratio ξ and natural frequency w_n are

$$\xi = \frac{B_c + C_{e0} + \Delta C_e}{2\sqrt{(K_c + G_{e0} + \Delta G_e)(M_c + M_{e0} + \Delta M_e)}}$$
$$w_n = \sqrt{\frac{K_c + G_{e0} + \Delta G_e}{M_c + M_{e0} + \Delta M_e}}. \tag{7}$$

Although M_{e0}, C_{e0}, G_{e0} and $\Delta M_e, \Delta C_e, \Delta G_e$ are determined by interactive environment, we can improve transient performances of the system (e.g., peak time, maximum overshoot, rise time) by tunning the impedance parameters M_c, B_c, K_c to affect damping ratio ξ and natural frequency w_n.

In this study, we focus on the rapid adaptive tuning strategy of the three impedance control parameters in response to dynamically evolving interactive environments. To efficiently obtain the suitable impedance control parameters for the current interactive object, a Bayesian optimization-based adaptive learning method is employed, as shown in the lower part of Fig. 1. In this method, three core steps are involved. First, a cost function derived from the real-time force feedback is designed to evaluate the performance of the robot interacting with the target environment under the current impedance control parameters. Second, a Gaussian process (GP) model, as the surrogate model of the cost function, is introduced to find an approximation of a complex mapping from the impedance control parameters to the specified cost function. The GP model serves as a prior probability distribution over the space of all feasible cost functions and is updated with each new measurement to yield a more accurate posterior distribution. In the final step, an acquisition function is chosen, which guides the optimization by determining the next potential impedance parameters to evaluate. The newly produced impedance parameters will be fed into the impedance control.

B. Bayesian Optimization-Based Efficient Adaptive Method

To find the suitable impedance control parameters, we need to construct a mapping from the impedance control parameters to the contact force between the robot and the interactive object. As stated above, a GP model is used to construct this mapping. The GP model is a commonly used non-parametric model and has been widely used in the fields of regression, classification, and black box function inference [22]. It has not only high flexibility but also can provide uncertainty estimation for the surrogate model. The GP model is defined as

$$S_t(x) \sim \mathrm{GP}\big(v(x), p\big(x, x'\big)\big) \tag{8}$$

where S_t is the specified cost function calculated by formula (4), $x = [M_c, B_c, K_c]^T$, $v(x)$, and $p(x, x')$ are

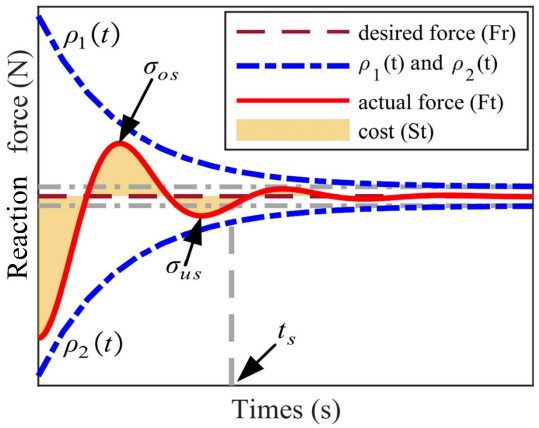


Fig. 2. Performance metrics used to describe the force curve and the iteration termination criterion.

mean function and covariance function, respectively. For convenience of calculation, the data are usually preprocessed, that is, set the mean function $v(x) = 0$.

Selecting a reasonable covariance function is key for accurately employing the GP. In our case, the robotic system periodically contacts the environment and impulsive dynamics are inevitably involved. Consequently, the widely used squared exponential covariance function is probably, in our opinion, not appropriate, because the squared exponential covariance function is infinitely differentiable, and the constructed GP is very smooth [23]. With these in mind, the ardsquaredexponential (ARD) Matérn 5/2 covariance function is finally chosen, taking into account the fact that: 1) the ARD Matérn 5/2 covariance function is not infinitely differentiable but has at least 2nd-order differentiability [24] and 2) the suitability of the ARD Matérn 5/2 covariance function for modeling complex physical processes has been demonstrated in some research [25], [26].

The expression of the covariance function is as follows:

$$p\big(x_i, x_j|\theta\big) = \sigma_f^2\left(1 + \sqrt{5}r + \frac{5}{3}r^2\right)\exp\left(-\sqrt{5}\ r\right) \tag{9}$$

where $r = (\sum_{m=1}^{d}((x_{im} - x_{jm})^2)/(\sigma_m^2))^{1/2}, \sigma_m$ is the characteristic length scale for each predictor $m \in \{1, \ldots, d\}$ and σ_f^2 is the signal variance.

In addition to the covariance function, the acquisition function, responsible for determining the next most potential control parameters based on the constructed GP model, should also be deliberately chosen, because it may influence the optimization capability and efficiency of BO. Among the commonly used probability of improvement (PI) [27], expected improvement (EI) [28], and lower confidence bound (LCB) acquisition functions [29], the EI algorithm is supposed to be more advantageous [30]. However, EI is also prone to fall into local optimum especially in the face of multipeaked surrogate models like the case in this article. To mitigate this weakness, the EI plus (EIP), an improvement of the EI algorithm to avoid local optima by modifying the kernel function [31], [32], is eventually adopted.

Another main component in Bayesian optimization is the cost function, which is used to quantify the difference in the performance of a closed-loop system controller when running

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

with different control parameters. To comprehensively reflect the performance characteristics of the force curve, the sum of the enclosed areas between the desired and the actual interaction force is adopted (shown in Fig. 2), which provides an intuitive identification regarding how much area is in a departure from the reference force. The calculation formula is

$$S_t = \sum_{t=t_0}^{t_e} | F_t - F_r | \cdot \text{step_size} \tag{10}$$

where F_t and F_r are the actual interaction force and desired force returned in each step-size, respectively; t_0 is the time of first contact with the environment, t_e defines the period during which the force data would be considered.

The pseudo-code of the proposed control strategy is given in Algorithm 1. The overall strategy can be decomposed into two distinct phases: random sampling and Bayesian optimization. In the random sampling phase, the control system executes ten randomized trials within the predefined impedance control parameter space and records the data during each iteration. Then, the Bayesian optimization is implemented. At this phase, the program first builds the GP model based on the data $D_{1:10}$ during the sampling process, and then selects the next group impedance parameters to evaluate x_{t+1} according to the acquisition function EIP. The entire BO phase will continue until the contact force curve converges within the region bounded by the prescribed performance function.

Algorithm 1 Bayesian Optimization-Based Impedance Controller Tuning

Input: environmental stiffness k_G, environmental damping d_G, distance from the reference plane h_G, impedance parameter ranges.
Output: Optimal solution of impedance control parameters (K_c, B_c, M_c)

```
for t = 1t = 1 to 10 do //random sampling
    Randomly sampled impedance control parameters K_t, B_t, M_t
    Implement impedance control with the robotic system, and
    calculate the cost S_t
end for // collecting the sampling data D_1:10
BO:
Construct initial GP model using sampling data D_1:10
repeat // Bayesian optimization
    Selects next group impedance parameters to evaluate
    according to the acquisition function
    Implement impedance control, calculate the cost S_t
    Update GP model
until the force curve satisfies the optimization criterion
```

C. Optimization Criterion and Performance Characterization

For the robot-environment interaction in this article, both steady-state and transient performances (i.e., overshoot and convergence speed) are considered during optimization implementation. The steady-state performance characterizes the steady-state error of the contact force curve, while reduced overshoot and fast convergence speed are helpful in reducing the impact on contact. Inspired by the prescribed performance control strategy [33], an optimization criterion consisting of two prescribed performance functions is designed. As shown in Fig. 2, the prescribed performance functions are as follows:

$$\rho_i(t) = \rho_i(0)^* \exp(-l_i t) + \rho_i(\infty)^* + F_r, \quad i = 1, 2 \tag{11}$$

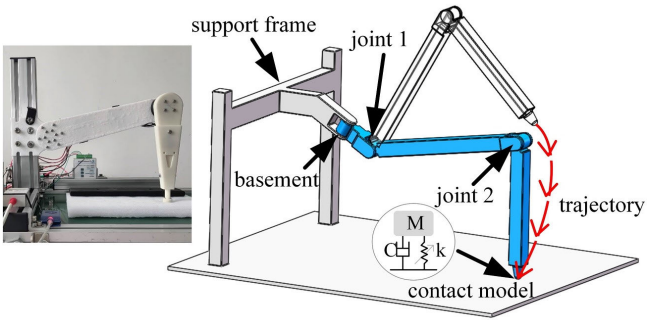


Fig. 3. Setup of the robotic manipulator simulation platform.

where $\rho_i(0)^*$, $\rho_i(\infty)^*$, and l_i denote the initial value, steady-state error value, and decay rate of the prescribed performance function, respectively. By adjusting the three parameters $\rho_i(0)^*$, $\rho_i(\infty)^*$, and l_i, the convergence speed and steady-state error of the interaction force can be constrained.

To systematically characterize the performance with different impedance control parameters, in addition to the enclosed area S_t that is employed as the cost function of BO, three transient metrics are further adopted, which are overshoot σ_{os}, undershoot σ_{us}, settling time t_s, as shown in Fig. 2. Furthermore, according to our experimental observations, there may exist an overdamped phenomenon in the contact force curve, i.e., there is no overshoot and undershoot. To tackle this, for the sake of unity, we take the force errors at the moments of 0.2 and 0.4 s after the robot interacts with the environment to replace overshoot σ_{os} and undershoot σ_{us}, respectively, and name them $\hat{\sigma}_{os}$ and $\hat{\sigma}_{us}$, respectively.

III. Simulation Study

To explore the performance of the proposed method in detail, we employ a two-DOF robotic manipulator interacting with table tops of different physical properties. Although the manipulator is simple, it allows us to implement a systematic investigation and evaluation of the proposed Bayesian optimization-based strategy, without being distracted by, for example, proportion integration differentiation (PID) parameter tuning and dynamics identification in multi DOF robotic systems that may also be important and influential for the performance of the robotic interaction system. At the same time, to highlight its effectiveness and efficiency, we compare the performance of the proposed control strategy against two benchmarks, genetic algorithm-based adaptive impedance control (benchmark 1) and indirect adaptive impedance control (benchmark 2) [11], [12].

A. Simulation Setup

The simulation platform, as depicted in Fig. 3, consists of a two-DOF robotic manipulator, a support frame of the robotic manipulator, and a material board with adjustable mechanical properties and geometric heights. The manipulator morphology is designed to resemble that of a planar two-link mechanism, consisting of two segments connected by joints. From proximal to distal, the lengths are 230 and 185 mm, respectively. The step size of the simulation experiment is set to 1 ms.

As illustrated in Fig. 3, during simulation implementations, the robotic manipulator is commanded to repeatedly touch

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

the material board with a certain trajectory to simulate the interactive motion of a robotic manipulator in actual working situations. An end-effector-environment contact model based on spring-damping is established to analyze the interaction process between the robotic manipulator and the material board. When the robotic manipulator contacts the board, a reaction force will be generated on the end-effector to balance the gravity and inertial forces of the system. The interaction process between the robotic manipulator and the board can be expressed by the following formula:

$$f_G = \begin{cases} k_G(z + h_G) + d_G\dot{z}, & z + h_G \geq 0 \text{ (in contact)} \\ 0, & z + h_G < 0 \text{ (not in contact)} \end{cases} \tag{12}$$

where f_G is the reaction force, the material board stiffness is k_G, the material board damping d_G, and h_G is the distance between the actual and the reference material board, z and $\dot{z}$ the position and velocity of the end-effector in the vertical direction, respectively. During simulation, k_G, d_G, and h_G can be modified to construct different material boards and interaction conditions. To enhance compatibility with the interacting object, a fixed-position material board, we omit the mass matrix from the environmental model in formula (1).

To verify the adaptive capability of the three different control methods under different material boards, four interactive objects are constructed. The specific parameters of these four material boards and the corresponding optimized range of impedance control parameters are shown in Table I. The constructed material boards vary in mechanical properties (e.g., stiffness and damping) and geometric heights. In addition, the optimization range of impedance control parameters is set according to the following rules:

1) The stability of the robotic manipulator system must be maintained.
2) Under the premise of satisfying the first rule, expand the final optimization scope as much as possible.

Two common adaptive impedance control methods are selected in this article as benchmarks for our approach, and the relevant settings of these two benchmark methods are explained as follows. For benchmark method 1, the initial population size is 30, and the impedance control parameters are optimized in the same range as our method. For benchmark method 2, initial estimate of the environmental position $\hat{x}_e$ and the environmental stiffness $\hat{k}_e$ corresponding to the four material board parameters are shown in Table I. For $\hat{x}_e$ and $\hat{k}_e$, the rules for determining the values of their initial estimates are the same as those for determining the range of impedance control parameters. In addition, the desired force in the simulation experiment is set to 5 N. At the same time, the parameters of the two prescribed performance functions are $\rho_1(0)^* = -\rho_2(0)^* = 3$, $l_1 = l_2 = 15$, $\rho_1(\infty)^* = -\rho_2(\infty)^* = 0.0005$, and $F_r = 5$.

B. Adaptation Efficiency

The number of learning times (the number of prior samples plus the number of iterations) required for the three adaptive impedance control methods for four material boards is summarized in Fig. 4. From an overall perspective, we can see from the figure that the Bayesian optimization-based adaptive strategy proposed in this article can obtain satisfied impedance control parameters after less than about 30 times of learning. However, the traditional adaptive impedance control strategies require hundreds of times of learning to obtain the corresponding impedance control parameters. The simulation results show that our proposed method has significant improvement in efficiency compared with the benchmark methods.

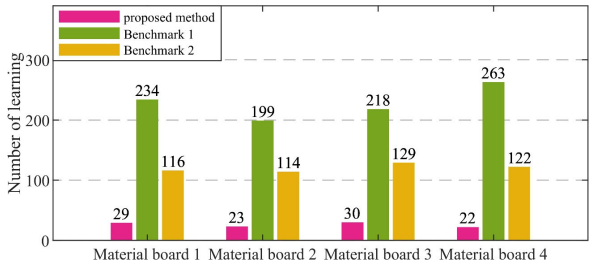


Fig. 4. Comparison of the learning times corresponding to three different control strategies for four material boards.

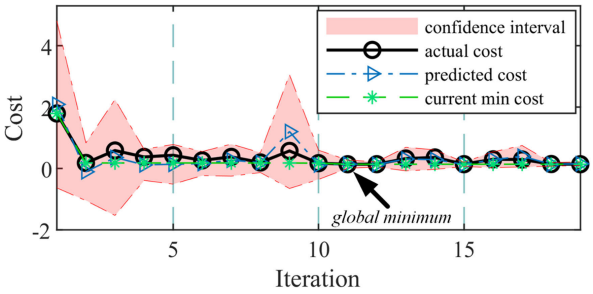


Fig. 5. Predicted, actual cost, and associated confidence interval at three standard deviations for our method.

Taking the simulation results from Material 1 as an example, we analyze the trend of cost values of the proposed algorithm during optimization, as shown in Fig. 5. During the initial 11 iterations of the optimization process, the predicted cost values exhibit high uncertainty due to the limited prior samples for constructing the initial surrogate model. However, as the number of iterations increases, the uncertainty of the predicted cost values gradually decreases, indicating that the surrogate model is getting closer to the real model in some feasible domains. This phenomenon can be seen after the 11th iteration when the predicted cost values are closer to the actual cost value than the previous 11 iterations. In addition, we can also know from the figure that the actual cost value of the second iteration is lower than the actual cost value of the third to the eighth iteration, which is caused by the BO algorithm choosing some evaluation points with high uncertainty in order to avoid the local optimum. Thus, the confidence interval for the third iteration is significantly larger than the confidence interval for the second iteration. Then, the cost values of the third to the eighth iteration show another downward trend. From the current minimum cost curve, the actual cost of the 11th iteration is the smallest in the entire iteration process, but the iteration is not stopped at this time, because the iteration termination criterion requires that the final optimized force curve must satisfy three predefined evaluation metrics, that is, convergence speed, overshoot, and steady-state error.

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

TABLE I

FOUR MATERIAL BOARD PARAMETERS FOR ALGORITHM VALIDATION AND CORRESPONDING OPTIMIZED RANGE OF IMPEDANCE CONTROL PARAMETERS. ESTIMATED VALUES FOR ENVIRONMENTAL POSITION AND ENVIRONMENTAL STIFFNESS

Material board	Material board parameters			Optimized range of impedance parameters			Parameter estimation for benchmark 2	
	$k_G\ [N/m]$	$d_G[Ns/m]$	$h_G[m]$	$M_C[kg]$	$B_C[Ns/m]$	$K_C[N/m]$	$\hat{x}_e[m]$	$\hat{k}_e[N/m]$
Board 1	10000	100	0.01	[1,5]	[50,500]	[500,1500]	0.004	9000
Board 2	10000	150	0.015	[1,5]	[50,500]	[600,2000]	0.007	9000
Board 3	50000	100	0.015	[1,5]	[50,1500]	[2500,8000]	0.007	40000
Board 4	50000	150	0.01	[1,5]	[50,2000]	[2500,7500]	0.004	40000

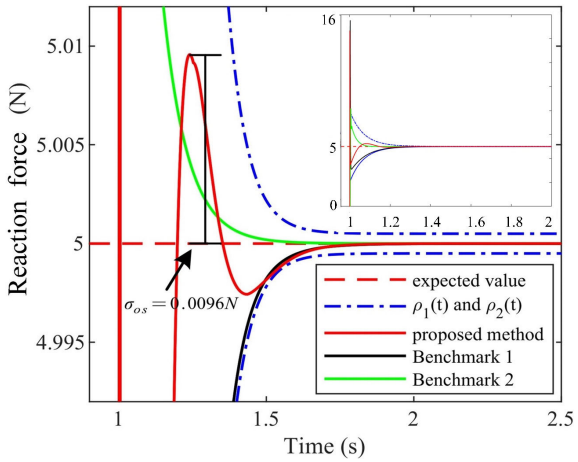


Fig. 6. Resultant force curves obtained from the three control strategies for material board 1.

C. Adaptation Performance

In the last section, we focus on analyzing the adaptation efficiency of the three adaptive impedance control methods for different material boards. Next, we will analyze the resultant force curves obtained from the three control strategies in four aspects (i.e., overshoot σ_{os}, undershoot σ_{us}, settling time t_s, and cost S_t).

Take the simulation data of material board 1 as an example. As shown in Fig. 6, all three optimized force curves converge perfectly into the area enclosed by the prescribed performance functions ρ_1 and ρ_2. However, the three force curves differ slightly in specific transient performance (i.e., overshoot σ_{os}, undershoot σ_{us}, settling time t_s, and cost S_t). First, the force curve optimized by the benchmark method 1 does not show overshoot, but its convergence speed is lower than the other two methods. Second, the force curve optimized by benchmark method 2 exhibits better performance, but this is obtained after 116 iterations. At the beginning of the iterative process of benchmark method 2, this method is prone to the phenomenon of long convergence time of the end-effector contact force due to the underestimation of the environmental parameters. Although the proposed method shows overshoot, the overshoot is only 0.0096 N. In addition, the difference between the three force curves in terms of steady-state error is negligible. In summary, we can conclude that our method can obtain impedance control parameters suitable for the current environment within a small number of learning times and that the resultant performance is no less than that with the impedance control parameter obtained by the benchmark methods.

TABLE II

COMPARISON OF THE TRANSIENT PERFORMANCES OF FORCE CURVES OPTIMIZED BY THREE ADAPTIVE IMPEDANCE CONTROL METHODS FOR FOUR MATERIAL BOARDS

Material Board	Control method	overshoot $\hat{\sigma}_{os}[N]$	undershoot $\hat{\sigma}_{us}[N]$	settling time $t_s[s]$
Material 1	proposed method	0.0006	0.0022	0.615
	Benchmark 1	0.1352	0.0069	0.598
	Benchmark 2	0.0067	0.0006	0.412
Material 2	proposed method	0.0411	0.0008	0.446
	Benchmark 1	0.15	0.0064	0.5
	Benchmark 2	0.0062	0.0005	0.391
Material 3	proposed method	0.0033	0.0015	0.6667
	Benchmark 1	0.0047	0.0007	0.451
	Benchmark 2	0.0033	0.0004	0.344
Material 4	proposed method	0.0108	0.0016	0.5285
	Benchmark 1	0.0138	0.0001	0.4871
	Benchmark 2	0.0034	0.0006	0.424

Next, we perform a detailed comparison of the transient performances of the resultant force curves optimized by the three adaptive impedance control methods across four material boards. The results are summarized in Table II. For Material board 1, our method is much better than the other two methods in terms of overshoot between the other two methods from the view of undershoot, and slightly behind the other two methods in settling time. For Material board 2, the performance of our method is between the benchmark methods in all 3 terms of overshoot, undershoot, and settling time. For Material board 3, our method is slightly inferior to the two benchmark methods in undershoot and settling time, but equal to benchmark method 2 and better than benchmark method 1 in overshoot. For Material board 4, our method is also slightly inferior to the two benchmark methods in undershoot and settling time, but in between the two benchmark methods in overshoot. Overall, for different material boards, each method exhibits their own advantages in certain of the three measures and acceptable performance in other measures. Taking into account the obvious advantage in adaptation efficiency with our method (roughly 30 times versus hundreds of times, see Fig. 4), our method is much more competitive.

IV. EXPERIMENTAL STUDY

In order to further demonstrate the effectiveness and efficiency of the proposed control strategy, the following experiments have been carried out with the real-world robotic

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

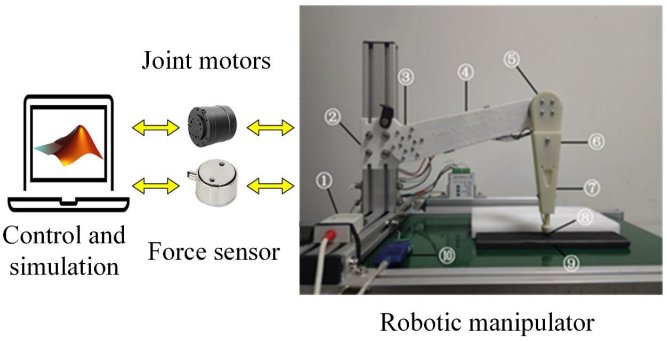


Fig. 7. Hardware setup of the robotic manipulator: ① CAN bus adapter, ② basement, ③ joint 1 motor, ④ link 1, ⑤ joint 2 motor, ⑥ link 2, ⑦ passive elastic cushioning mechanism, ⑧ force sensor, ⑨ material board, and ⑩ RS485 adapter.

manipulator. In this section, we will introduce the experimental setup, and then experimental results over various interactive conditions will be displayed and analyzed in detail.

A. Experimental Setup

The robotic manipulator designed in this article and its hardware configuration are shown in Fig. 7. We represent table tops with different physical properties by placing different material boards. The robotic manipulator consists of two rotary joints. Each joint is powered by a dc servomotor with an encoder for measuring the motor position. The end link is sequentially composed of, from top to bottom, a spring cushioning mechanism, a force sensor, and a semicircular rubber. After interacting with the material board, the actual contact force and the joint motor position information are fed back to the upper computer via the RS485 adapter and the controller area network (CAN) bus adapter. The upper computer outputs adjustment commands based on the contact force error, calculated by the impedance control algorithm, and sends them to the joint motor. Like the simulation runs, the robotic manipulator is commanded to repeatedly contact the boards in a specified way and impedance control parameters are iteratively adjusted.

To verify the performance of the proposed control strategy, experimental studies over two material boards are conducted: One is steel board (experiment one), and the other is foam board (experiment two), which represent typical stiff and soft materials the robotic manipulator may encounter. The robotic manipulator is set to contact the steel board and foam board with the desired interaction force of 20 N. In addition, the initial height of the end-effector from the material board is approximately 16 cm. At the same time, the parameters of the two prescribed performance functions are: $\rho_1(0)^* = 30$, $\rho_2(0)^* = -20$, $l_1 = l_2 = 4.5$, $\rho_1(\infty)^* = -\rho_2(\infty)^* = 0.4$, and $F_r = 5$. In the physical experiment, the principle of determining the optimization range of impedance control parameters is consistent with the principle of determining the optimization range of impedance control parameters in simulation.

B. Experimental Results

1) Experiment One: The material board used in this experiment is a steel board. For the stiff material, the optimization ranges of impedance control parameters are set to $M_c \in [1, 5]$, $B_c \in [30, 600]$, and $K_c \in [4500, 9500]$. The results of this experiment are shown in Fig. 8.

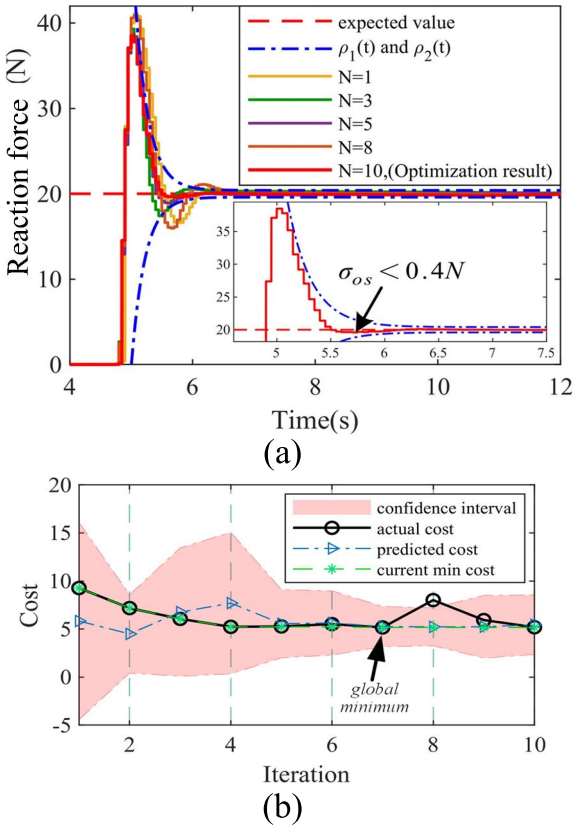


Fig. 8. (a) Interaction force over the steel board with the real-world robotic manipulator. (b) Cost trends during optimization.

In the experiment, the amplitude of force curve oscillation in the sampling phase is large. However, it can be seen from Fig. 8(a) that the amplitude of fluctuation of the force curve is relatively flat during the optimization process, which indicates that the impedance control parameters from our method are gradually approaching the feasible domain that satisfies the prescribed requirements. The final optimized force curve has an overshoot of no more than 0.4 N and converges perfectly to the prescribed region. The impedance control parameters that finally meet the specified requirements are $M_c = 3.53$ kg, $B_c = 30.08$ N/(m/s), and $K_c = 9486.46$ N/m.

Fig. 8(b) represents the trend of actual cost, predicted cost, and confidence interval during the optimization process. As can be seen from the figure, the actual cost gradually decreases as the number of iterations increases. Among them, the actual cost of the eighth iteration appears to increase, which is due to the low number of samples used for modeling in the early stage of the optimization process, so the modeling accuracy in unexplored regions is not enough. In Bayesian optimization, the algorithm tries to explore these unknown areas and may select the point whose predicted cost is minimal in these areas for subsequent sampling. As a result, the predicted cost differs from the actual cost due to the disparity between the surrogate model and the true underlying mapping. In addition, the predicted cost, and the actual costs at first, second, and fourth iteration are also quite different. As the modeling accuracy increases, the cost after the eighth iteration

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

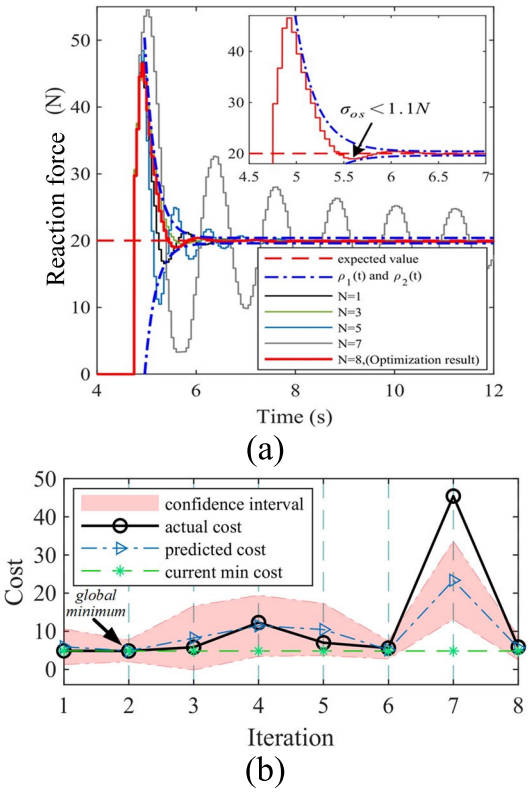


Fig. 9. (a) Interaction force over the foam board with the real-world robotic manipulator. (b) Cost trends during optimization.

gradually decreases and the predicted cost gradually matches the actual cost. In order to have excellent dynamic and stable performance of the final optimized force curve, the iterations are not stopped even if the actual cost value of the seventh iteration is the lowest in the whole optimization process.

2) Experiment Two: In this experiment, we use the foam board as an interactive object. The optimization ranges of impedance control parameters are set to $M_c \in [1, 5]$, $B_c \in [20, 500]$, and $K_c \in [2500, 5000]$. The results of this experiment are shown in Fig. 9.

It can be seen from Fig. 9(a) that the force curve during optimization is relatively flat. Although the force curve of the seventh iteration fluctuates considerably, this is due to the large range of the optimization performed, so there are occasional evaluation points that cause the force curve to fluctuate. The final impedance control parameters that meet the specified requirements are $M_C = 1.05$ kg, $B_c = 20.72$ N/(m/s), and $K_c = 4479.32$ N/m.

Fig. 9(b) describes the trend of cost in the optimization process. Corresponding to the force curve of the seventh iteration in Fig. 9(a), an extreme value of the cost in the optimization process appears, i.e., the cost of the seventh iteration. The reason for the observed fluctuations in the cost value is consistent with what was noted in Section IV-B1. However, this does not adversely impact the overall performance of the algorithm. Throughout the optimization process, the trends of the predicted cost values and the actual cost values largely coincide, which indicates that the accuracy of the constructed model is relatively high around these evaluation points. Therefore, the number of learning iterations is also less.

TABLE III
ANALYSIS OF THE TRANSIENT PERFORMANCES OF THE RESULTANT FORCE CURVES IN BOTH STEEL AND FOAM BOARD

Material board	overshoot $\hat{\sigma}_{os}[N]$	undershoot $\hat{\sigma}_{us}[N]$	settling time $t_s[s]$
Steel	11.5	0.46	0.65
Foam	15.86	1.48	1.251

Similarly, we analyze the transient performances of the optimized force curves in the physical experiments. According to our experimental observations, we take the force error values corresponding to the moments of 0.3 and 0.6 s after the robotics manipulator touches the material board as the estimated values of overshoot and undershoot, respectively. The results of the analysis are shown in Table III. It can be seen that, the overshoots over the steel and foam boards are 11.5 and 15.86 N, but with, respectively, 0.65 and 1.251 s of settling, the force curves have lain in the specific steady-state zone, indicating that the impedance control parameters found by our method enable the contact force of the robotics manipulator to drop quickly to near the desired force. In addition, it can be seen that the three measures over the soft foam board are relatively larger than those over the steel board. This observation is consistent with the fact that preciously interacting with soft materials is, compared to rigid body, much complicated due to the inherent nonlinear deformation.

V. DISCUSSION AND CONCLUSION

In this article, we have investigated the Bayesian optimization-based adaptive impedance control, for properly and efficiently tuning the inertial, damping, and stiffness parameters of impedance control according to the real time force feedback, thereby improving the force control performance of robotic interactive tasks. With a two DOF robotic manipulator built into both numerical model and real-world prototype, we systematically introduce the implementation and tuning process of the proposed method. Both numerical and experimental results indicate that proper parameters of impedance control can be obtained with as few as a dozen interaction iterations, which is expected to improve the real-world applications of robotic manipulators under changing interactive conditions.

A. Adaptation Robustness Against Prior Data

In the implementation of adaptive impedance control, the construction of the GP model based on the prior data is a key, upon which the model can be further updated with observed posteriors. It is obvious that excellent priors would reduce the scope of the optimal solution, making the acquisition function search for the candidate solution in a smaller range, thereby reducing the probability of unnecessary points and thus the interaction effort of the robotic manipulator. In this work, ten random samples of impedance parameters along with the corresponding interaction force data are employed as the priors. To clarify the robustness of the proposed Bayesian adaptive method against prior data, a supplementary analysis is undertaken. As shown in Fig. 10(a) the results

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

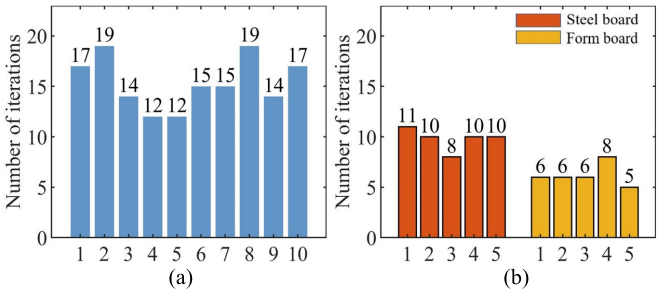


Fig. 10. Number of iterations corresponding to different prior samples in (a) simulation and (b) physical experiments. (The initial number of samples for each experiment is ten.)

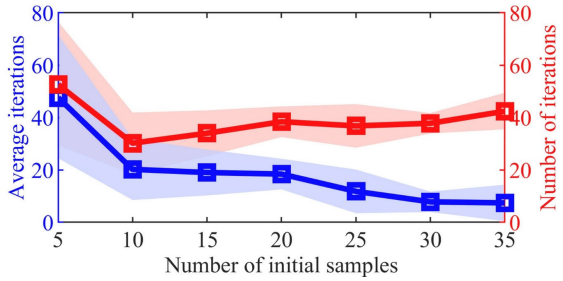


Fig. 11. Number of iterations corresponding to the number of different initial sample points.

are obtained with ten groups of different initial sample data, under the simulation setting in Section III. As can be seen, despite different initial sample data, our method can produce suitable impedance parameters via 15.5 $\pm$ 3.5 interaction iterations, exhibiting good robustness against prior data. The robustness against prior data is also verified by physical experiments, in which steel and foam boards are used and five experimental runs with different initial sample data are implemented on each material board. The experimental results are shown in Fig. 10(b), where the fluctuations in the number of optimization iterations in each material board are below 3, indicating that the proposed algorithm has good robustness.

B. Toward the Number of Prior Data

Ten prior samples were utilized in this study. Additionally, we explore the effect of different numbers of prior data on adaptation efficiency. Since we are trying to find the impedance control parameters that meet the convergence conditions more quickly, we limit the number of initial samples to 35 and conduct investigations at an interval of 5. The simulation environment used in this exploration is the same as the settings for interacting with material board 1 in Section III-B. We conducted five experiments for each number of initial samples and averaged the number of iterations in the optimization process, as shown by the blue line in Fig. 11.

Fig. 11 reveals that the number of iterations in the optimization process gradually decreases as the initial number of samples increases, indicating a progressive enhancement in the optimization efficiency of the control strategy. This is obvious because the initial surrogate model with more prior knowledge is more advantageous for the optimization search. However, the optimization efficiency improves little after the initial number of samples reaches 30. The red line in Fig. 11 shows the total number of adaptations, that is the number of iterations in the optimization process plus the number of initial samples. From the trend of this line, the total number of adaptations tends to decrease and then increase as the number of initial samples increases. This indicates that it is not favorable to have a larger number of initial samples from the perspective of overall adaptation efficiency. Conversely, since the initial samples are randomly sampled in the feasible domain, there may be random samples that cause large fluctuations in the interaction force between the robotic manipulator and the interactive object, which is not expected. Furthermore, the optimization process is adaptively executed by continuously assessing performance and initiating resampling and reoptimization if the interactive condition changes. In this regard, the total number of adaptations seems a crucial factor in maintaining adaptation efficiency.

C. Optimization Ranges and Future Work

In the process of searching impedance control parameters, the first thing to consider is to maintain the stability of the entire closed-loop control system. To achieve this, the search ranges of each impedance control parameter are manually given in advance, based on our knowledge from operating the system. This is a straightforward but conservative way. In fact, the ranges of the three control parameters should be coupled and varied as interactive conditions. Also, it is tricky to find an explicit mapping from the impedance control parameters to system stability. This issue could be further solved by using constrained Bayesian optimization, where another GP model to estimate system stability is added and used in the Bayesian inference [34], [35].

REFERENCES

[1] G. Du, Y. Deng, C. Li, and P. X. Liu, "A mobile natural interactive technique with bare hand manipulation and unrestricted force feedback for virtual assembly tasks," *IEEE Trans. Instrum. Meas.*, early access, Feb. 4, 2022, doi: 10.1109/TIM.2022.3149087.

[2] X. Bao, S. Wang, L. Zheng, R. J. Housden, J. V. Hajnal, and K. Rhode, "A novel ultrasound robot with force/torque measurement and control for safe and efficient scanning," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023, doi: 10.1109/TIM.2023.3239925.

[3] T. Yoshiike et al., "Development of experimental legged robot for inspection and disaster response in plants," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2017, pp. 4869–4876, doi: 10.1109/IROS.2017.8206364.

[4] M. J. Pollayil et al., "Choosing stiffness and damping for optimal impedance planning," *IEEE Trans. Robot.*, vol. 39, no. 2, pp. 1281–1300, Apr. 2023, doi: 10.1109/TRO.2022.3216078.

[5] S. Wolf et al., "Variable stiffness actuators: Review on design and components," *IEEE/ASME Trans. Mechatronics*, vol. 21, no. 5, pp. 2418–2430, Oct. 2016, doi: 10.1109/TMECH.2015.2501019.

[6] N. Hogan, "Impedance control: An approach to manipulation," in *Proc. Amer. Control Conf.*, Jul. 1984, pp. 304–313, doi: 10.23919/ACC.1984.4788393.

[7] A. Albu-Schaffer, C. Ott, U. Frese, and G. Hirzinger, "Cartesian impedance control of redundant robots: Recent results with the DLR-light-weight-arms," in *Proc. IEEE Int. Conf. Robot. Autom.*, Jun. 2003, pp. 3704–3709, doi: 10.1109/ROBOT.2003.1242165.

[8] L. Liu, M. Illian, S. Leonhardt, and B. J. E. Misgeld, "Iterative learning control for cascaded impedance-controlled compliant exoskeleton with adaptive reaction to spasticity," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023, doi: 10.1109/TIM.2023.3286004.

[9] K. Xu et al., "Adaptive impedance control with variable target stiffness for wheel-legged robot on complex unknown terrain," *Mechatronics*, vol. 69, Aug. 2020, Art. no. 102388, doi: 10.1016/j.mechatronics.2020.102388.

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.

[10] A. Taherifar, G. R. Vossoughi, and A. Ghafari, "Variable admittance control of the exoskeleton for gait rehabilitation based on a novel strength metric," *Robotica*, vol. 36, pp. 1–21, Nov. 2017, doi: 10.1017/S0263574717000480.
[11] Y.-G. Zhu, B. Jin, and W. Li, "Leg compliance control of a hexapod robot based on improved adaptive control in different environments," *J. Central South Univ.*, vol. 22, no. 3, pp. 904–913, Mar. 2015, doi: 10.1007/s11771-015-2600-0.
[12] J. Duan, Y. Gan, M. Chen, and X. Dai, "Adaptive variable impedance control for dynamic contact force tracking in uncertain environment," *Robot. Auto. Syst.*, vol. 102, pp. 54–65, Apr. 2018, doi: 10.1016/j.robot.2018.01.009.
[13] T. Tsuji, K. Ito, and P. G. Morasso, "Neural network learning of robot arm impedance in operational space," *IEEE Trans. Syst., Man, Cybern., B Cybern.*, vol. 26, no. 2, pp. 290–298, Apr. 1996, doi: 10.1109/3477.485879.
[14] F. Stulp, J. Buchli, A. Ellmer, M. Mistry, E. A. Theodorou, and S. Schaal, "Model-free reinforcement learning of impedance control in stochastic environments," *IEEE Trans. Auto. Mental Develop.*, vol. 4, no. 4, pp. 330–341, Dec. 2012, doi: 10.1109/TAMD.2012.2205924.
[15] F. Winter, M. Saveriano, and D. Lee, "The role of coupling terms in variable impedance policies learning," in *Proc. Int. Workshop Hum.-Friendly Robot.*, Genoa, Italy, 2016.
[16] Y. Hu, X. Wu, P. Geng, and Z. Li, "Evolution strategies learning with variable impedance control for grasping under uncertainty," *IEEE Trans. Ind. Electron.*, vol. 66, no. 10, pp. 7788–7799, Oct. 2019, doi: 10.1109/TIE.2018.2884240.
[17] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, "Taking the human out of the loop: A review of Bayesian optimization," *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, Jan. 2016, doi: 10.1109/JPROC.2015.2494218.
[18] R. Calandra, A. Seyfarth, J. Peters, and M. P. Deisenroth, "An experimental comparison of Bayesian optimization for bipedal locomotion," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Jun. 2014, pp. 1951–1958, doi: 10.1109/ICRA.2014.6907117.
[19] J. Li, R. Chen, X. Huang, and Y. Qu, "Development of deep residual neural networks for gear pitting fault diagnosis using Bayesian optimization," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–15, 2022, doi: 10.1109/TIM.2022.3219476.
[20] Y. Li, S. S. Ge, and C. Yang, "Impedance control for multi-point human–robot interaction," in *Proc. 8th Asian Control Conf. (ASCC)*, 2011, pp. 1187–1192.
[21] J. Chen, Y. Liu, J. Zhao, H. Zhang, and H. Jin, "Biomimetic design and optimal swing of a hexapod robot leg," *J. Bionic Eng.*, vol. 11, no. 1, pp. 26–35, 2014, doi: https://doi.org/10.1016/S1672-6529(14)60017-2.
[22] L. Jin, J. Rückin, S. H. Kiss, T. Vidal-Calleja, and M. Popovic, "Adaptive-resolution field mapping using Gaussian process fusion with integral kernels," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 7471–7478, Jul. 2022, doi: 10.1109/LRA.2022.3183797.
[23] C. Rasmussen, O. Bousquet, U. Luxburg, and G. Rätsch, "Gaussian processes in machine learning," in *Advanced Lectures on Machine Learning*, vol. 3176. Berlin, Germany: Springer, Sep. 2004, doi: 10.1007/978-3-540-28650-9_4.
[24] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," in *Proc. 25th Int. Conf. Neural Inf. Process. Syst.*, vol. 2. Red Hook, NY, USA: Curran Associates, 2012, pp. 2951–2959.
[25] H. Huang, Z. Li, X. Peng, S. X. Ding, and W. Zhong, "Gaussian process regression with maximizing the composite conditional likelihood," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021, doi: 10.1109/TIM.2021.3104376.
[26] M. Maier, R. Zwicker, M. Akbari, A. Rupenyan, and K. Wegener, "Bayesian optimization for autonomous process set-up in turning," *CIRP J. Manuf. Sci. Technol.*, vol. 26, pp. 81–87, Aug. 2019, doi: 10.1016/j.cirpj.2019.04.005.
[27] X. Ruan, P. Jiang, Q. Zhou, J. Hu, and L. Shu, "Variable-fidelity probability of improvement method for efficient global optimization of expensive black-box problems," *Structural Multidisciplinary Optim.*, vol. 62, no. 6, pp. 3021–3052, Dec. 2020, doi: 10.1007/s00158-020-02646-9.
[28] D. R. Jones, M. Schonlau, and W. J. Welch, "Efficient global optimization of expensive black-box functions," *J. Glob. Optim.*, vol. 13, no. 4, pp. 455–492, Dec. 1998, doi: 10.1023/A:1008306431147.
[29] E. Contal, D. Buffoni, A. Robicquet, and N. Vayatis, "Parallel Gaussian process optimization with upper confidence bound and pure exploration," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*, Apr. 2013, pp. 225–240, doi: 10.1007/978-3-642-40988-2_15.
[30] J. Liu, Z.-H. Han, and W. Song, "Comparison of infill sampling criteria in Kriging-based aerodynamic optimization," in *Proc. 28th Congr. Int. Counc. Aeronaut. Sci. (ICAS)*, vol. 2, Jan. 2012, pp. 1625–1634.
[31] A. D. Bull, "Convergence rates of efficient global optimization algorithms," *J. Mach. Learn. Res.*, vol. 12, pp. 2879–2904, Oct. 2011.
[32] W. Lyu et al., "An efficient Bayesian optimization approach for automated optimization of analog circuits," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 6, pp. 1954–1967, Jun. 2018, doi: 10.1109/TCSI.2017.2768826.
[33] Z. Doulgeri and Y. Karayiannidis, "PID type robot joint position regulation with prescribed performance guaranties," in *Proc. IEEE Int. Conf. Robot. Automat.*, Jul. 2010, pp. 4137–4142, doi: 10.1109/ROBOT.2010.5509676.
[34] M. Khosravi, C. König, M. Maier, R. S. Smith, J. Lygeros, and A. Rupenyan, "Safety-aware cascade controller tuning using constrained Bayesian optimization," *IEEE Trans. Ind. Electron.*, vol. 70, no. 2, pp. 2128–2138, Feb. 2023, doi: 10.1109/TIE.2022.3158007.
[35] C. Yu, J. Cao, and A. Rosendo, "Learning to climb: Constrained contextual Bayesian optimisation on a multi-modal legged robot," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 9881–9888, Oct. 2022, doi: 10.1109/LRA.2022.3192798.



Jie Chen received the M.S. and Ph.D. degrees in mechanical engineering from the Harbin Institute of Technology, Harbin, China, in 2012 and 2017, respectively.

From 2014 to 2015, he was a Visiting Doctoral Student at the University of Cambridge, Cambridge, U.K. He is currently an Associate Professor with Northeastern University, Shenyang, China. His research interests include robot design and control.



Linsong Deng received the B.S. degree in mechanical engineering from Dalian Jiaotong University, Dalian, China, in 2020. He is currently pursuing the M.Sc. degree in mechanical engineering and automation with Northeastern University, Shenyang, China.

His research interests include mobile robots and motion control.



Zhengchun Hua received the B.S. degree in mechanical engineering and automation from Northeastern University, Shenyang, China, in 2021, where he is currently pursuing the M.Sc. degree in mechanical engineering and automation.

His research interests include mobile robots and motion control.



Weiyi Ying received the B.S. degree in mechanical engineering from Northeastern University, Shenyang, China, in 2022, where he is currently pursuing the M.Sc. degree in mechanical engineering and automation.

His research interests include mobile robots and motion control.



Jie Zhao (Senior Member, IEEE) received the B.S. and Ph.D. degrees in mechatronics engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 1990 and 1996, respectively.

He is currently a Professor with the State Key Laboratory of Robotics and System, Harbin University of Technology. His research interests include industrial robots and bionic robots.

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on October 28,2026 at 14:49:01 UTC from IEEE Xplore. Restrictions apply.